Abstract

The parent study ranks eight candidate shapes and recommends a single winner. This note does the opposite and stays inside the brand: it keeps the letter G and asks what the ring can be made to *mean*. Four variants are rendered with the production material, lighting, and 40 degree camera, and each is justified by a single geometric device that carries a single semantic channel. Every device is a change to the parameterisation, not to the artwork, so each variant remains a function rather than a mesh.

1 Design rule

Each variant must satisfy three constraints: it must be a closed or layer-stacked surface so that the object reads as solid rather than torn; it must be expressible with the existing two instanced primitive types and one recycled seam; and it must carry its meaning in *geometry*, so that the reading survives on mobile, where the production renderer disables bloom and depth of field. Motion is used only to confirm the reading, never to supply it.

2 The four variants

V1 — closed G torus: history has no free ends

Device. Append a chord from the inner terminal to the defocused head, resample the resulting closed polyline, and sweep the same three-row hexagonal sheet around it with $\varphi(u) \equiv 0$:

$$P(u, v) = \gamma(u) + r\, N(u), \qquad \gamma(0) = \gamma(1).$$

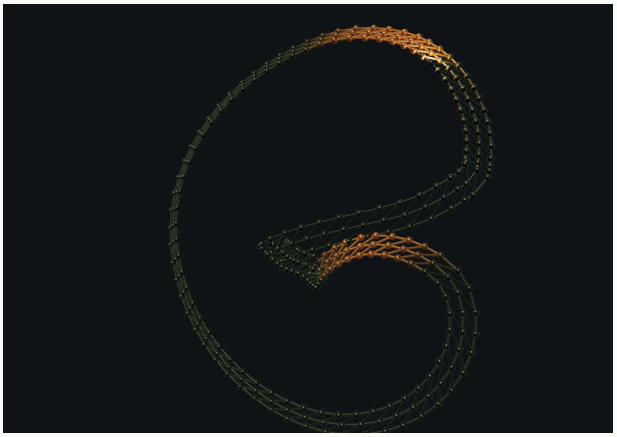
Semantics. The current shape is an open C whose two ends fade out. A fading end is a closure violation that a viewer reads as damage rather than as depth. Closing the loop converts the same silhouette into a solid body and, more importantly, makes one lap *mean* one period: the ring is now the domain convention for a cyclical quantity, and position along it is the most accurately decoded channel available. Two antipodal pulses replace the single travelling one, because a single pulse on a closed loop has no phase reference — it is traffic. A pair defines the period.
Cost. The chord across the aperture is not part of the logo and must be drawn as deliberate structure. Node and bond counts are unchanged.

V2 — phase-locked G torus: the period becomes geometry

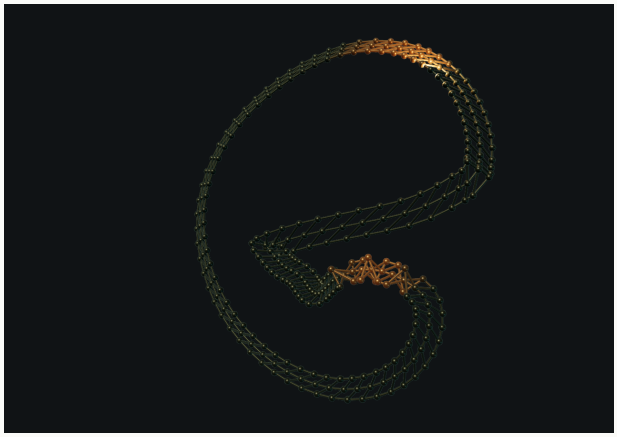
Device. Keep V1's closed centreline and replace the flat frame by the parallel-transported triple $\{T, N, B\}$, rotating the cross-section at constant rate:

$$\Phi_k(u, v) = \gamma(u) + r\Big(\cos\varphi(u)\, N(u) + \sin\varphi(u)\, B(u)\Big), \qquad \varphi(u) = \varphi_0 + 2\pi k u.$$

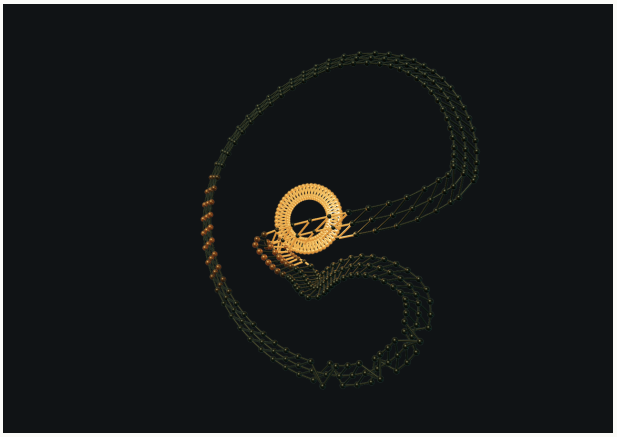
For integer k the cross-section returns to itself after one lap, so the surface still closes with a single seam.
Semantics. This is the variant the parent study recommends. The exponential winding makes k a geometric statement about the sampling period: one lap is one period, and k is the number of phase revolutions inside it. The motif passes from edge-on to face-on around the loop, which produces an internal light-to-dark gradient that survives at 40 px — exactly the legibility that the current shape borrows from post-processing and therefore loses on mobile. With the pulse pair running slightly off the twist rate, their beat frequency $f_{\text{beat}} = \left|1/T_{\text{p}} - k/T_{\text{t}}\right|$ becomes a legible picture of two phase-locked cycles, which is the product claim rendered as geometry.
Cost. The G tail doubles back, so the transported frame can spin where the curvature is extreme; the mitigation is to relax the tail's radius in the control points and to clamp $|d\varphi/ds| r < 0.6$.



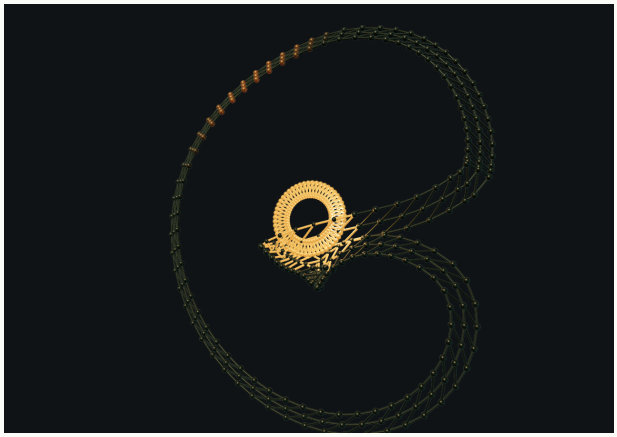
(a) **V1 — closed G torus.** The head and tail are joined by a chord across the aperture, so the loop has no free ends.



(b) **V2 — phase-locked G torus.** The cross-section rotates once per lap, $k = 1$, over a rotation-minimising frame.



(c) **V3 — composite.** V2's twisted closed shell with V4's core on the same axis; the shell carries the period, the core the conjunction.



(d) **V4 — concentric core.** A closed G shell around a small gold torus on the same axis.

Figure 1: The four variants, rendered with the production material, lighting, and camera. Gold marks the active signal; carbon is unlit.

V3 — composite: the recommended build

Device. Superimpose the two strongest devices without blending them: V2's helically twisted closed shell, and V4's inner torus on the same axis. The shell keeps $k = 1$ and its antipodal pulse pair; the core keeps a steady emission at 0.38 radius.
Semantics. The two channels stay separable, which is the point. The shell answers *when*: one lap is one period, k phase revolutions per period, two pulses defining phase. The core answers *what*: observations coaxial with, and inside, relationships. Because the core is a distinct body rather than a texture on the shell, the eye reads a two-part object, and the sentence *relationships and observations, together* becomes a statement about containment instead of a caption. The twist supplies an internal light-to-dark gradient, so the reading survives at small sizes and without post-processing.
Cost. Two assemblies in one draw path, and the core must stay inside the aperture at every rotation. In practice this is the cheapest version of the recommended design, because it reuses the two previous parameterisations unchanged.

A rejected variant, and why

An earlier fifth candidate, a *temporal stack* — five contours of the same G at increasing depth joined by rungs, one contour lit — is archived in `variations/archive/` rather than presented. Its semantics were the most literal of all: a layer is a snapshot, rungs are state transitions, depth is time. It was rejected on visual grounds, not semantic ones. Five superimposed contours destroy the silhouette, the rungs read as a grid rather than as causality, and the object no longer holds together as a body — which contradicts the one rule the rest of the set obeys. Recording the rejection matters: the failure is informative, because it shows that encoding time *as layers* competes with encoding time *as position*, and only one of them can own the silhouette.

V4 — concentric core: two data models, one axis

Device. Keep V1's closed shell at radius ~ 2 and add a small torus of radius 0.38 on the same axis, sharing the same material but permanently emissive, with the shell carrying the travelling pulse.
Semantics. This is the most literal reading of the positioning sentence. The outer lattice is relationships; the inner ring is observations; they are concentric and coaxial, so neither is privileged and the claim *relationships and observations, together* is a geometric fact rather than a caption. It also produces the strongest depth cue in the set — a genuinely occluded core — which is what makes the object read as photographed.
Cost. A second primitive assembly in one draw path, and the core must be small enough to stay inside the aperture at every rotation.

3 Comparison

4 Reproduction

```
cd site
npm run dev
PREVIEW_BASE=http://127.0.0.1:5174/graphfin \
PLAYWRIGHT_CHROMIUM_EXECUTABLE=... \
  node research/variations/capture.mjs
pdflatex hero-g-variations.tex
```

`research/variations/preview.js` builds each variant from the same logo skeleton used by the production hero, so the silhouette cannot drift from the brand mark. `?v=` selects a variant and `?z=` overrides the camera distance. The rejected temporal stack is retained in `variations/archive/` and is still reachable with `?v=stack`.

Variant	**Geometric device**	**Semantic channel**	**Cost**
V1 closed G torus	close the loop, $\varphi = 0$	one lap = one period; no dissolved ends	chord across the aperture must read as designed
V2 phase-locked	helical sweep, $k = 1$, transported frame	period and phase as geometry; beat frequency	frame spin at the doubled-back tail
V3 composite	twisted closed shell + inner core	shell carries the period, core the conjunction	two assemblies in one draw path
V4 concentric core	closed shell + inner torus	relations outside, observations inside, coaxial	second assembly; core must fit the aperture

Table 1: Each variant buys one clear channel and pays in exactly one place. V2 is the recommendation of the parent study; V4 is the strongest purely visual option; V1 is the cheapest and safest; V3 is the most literal about the data.